Innovative Strategies in Crafting High-Performance Hard Carbon Electrodes for Next-Gen Batteries

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Promising next-generation substitutes for the widely used lithium-ion batteries (LIBs) are sodium- and potassium-ion batteries. They still have a lower energy density than LIBs, though. Japanese researchers investigated a novel approach to transform hard carbon into a superior negative electrode material to address this problem.

They created nanostructured hard carbon during synthesis by using inorganic zinc-based compounds as a template. This material performs exceptionally well in both alternative batteries.

With a wide range of uses, lithium-ion batteries (LIBs) are by far the most popular kind of rechargeable batteries.

These include spacecraft, renewable energy systems, electric cars (like Tesla cars), and consumer electronics.

When compared to other rechargeable batteries, LIBs perform better in many areas, but they are not without drawbacks of their own.

Since lithium is a relatively rare resource, its price will increase as its supply decreases in the future.

Furthermore, because the liquid electrolytes used in LIBs are frequently toxic and flammable, both the extraction of lithium and the incorrect disposal of LIBs present significant environmental challenges.

Researchers from all over the world are searching for alternative energy storage technologies as a result of LIBs' shortcomings.

Potassium- and sodium-ion batteries (NIBs and KIBs) are two quickly developing, affordable, and environmentally friendly solutions.

By the end of the decade, both KIBs and NIBs are expected to be billion-dollar industries.

Furthermore, a lot of money is being invested in this technology by businesses like **Faradion Limited** , **Tiamat Energy** , and **HiNa Battery Technology Co., Ltd.**

It is anticipated that **BYD** and **CATL** will release NIB-equipped electric vehicle battery packs shortly.

Regretfully, though, the NIB and KIB electrode materials' capacity still falls short of that of LIBs.For NIBs and KIBs, a group of researchers has been developing novel high-capacity electrode materials.

The most recent study reports on a novel synthesis approach for nanostructured "hard carbon" (HC) electrodes with previously unheard-of performance. It was published in Advanced Energy Materials on November 9, 2023.

However, what is HC and how does it help KIBs and NIBs? In contrast to other carbon-based materials like graphene or diamond, HC is amorphous, meaning it lacks a distinct crystalline structure.

It is resistant and strong as well. Magnesium oxide (MgO) was utilized as a template to change the final nanostructure of the HC electrodes for NIBs in a study published in 2021.